MIL-A-11928B
17 JUNE 1960
SUPERSEDING
MIL-A-11928A(SigC)
18 NOVEMBER 1958

MILITARY SPECIFICATION

AXLE RL-27( ) (FOR LAYING FIELD WIRE)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the requirements for Axle RL-27( ) used for carrying, reeling and unreeling Reel DR-4 and Wire Reel RL-159/U.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, forms a part of this specification.

SPECIFICATIONS

FEDERAL

T-T-871 — Twine, Cotton, Wrapping.

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).

PPP-P-291 — Paperboard, Wrapping, Cushioning.

PPP-B-585 — Boxes; Wood, Wirebound.

PPP-B-591 — Boxes; Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant.

MILITARY

MIL-P-116 — Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U.S. Military Property.

MIL-STD-169 — Extreme Temperature Cycle.

MIL-STD-170 — Moisture - Resistance Test Cycle for Ground Signal Equipment.

FSC 3895

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DRAWINGS

SIGNAL CORPS

SC–C–33073 — Form to be Used for Pilot Back Process Sheet.

SC–DL–83170 — Axle RL–27–( ).

(Copies of specifications, standards, and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.).

3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed in accordance with Drawing SC–DL–83170. However, if specified performance characteristics cannot be obtained by strict conformance with the drawing, the specified performance characteristics shall govern.

3.2 Operation. Each Axle RL–27–( ) shall be combined with corresponding components and associated equipments to form a complete operational unit as required. Under the test conditions specified in 4.11 the equipment shall operate smoothly without binding or catching.

3.3 Interchangeability. Corresponding units and replaceable parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used (see 4.9). When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular part is to be considered replaceable, the contracting officer shall be consulted.

3.4 Service conditions. The equipment shall meet the following service conditions.

3.4.1 *Equipment nonoperating.* The equipment shall comply with the operational requirements of 3.4.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) *Temperature.* Exposure in the range of + 160° to — 80° F.; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time (see 4.7).

(b) *Relative humidity.* Relative humidity up to 97 percent for 20 hours and exposure at 100-percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions, the equipment shall meet the requirements of 3.4.2.

(c) *Bounce and shock.* As described in 3.5.

3.4.2 *Equipment operating.* The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

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(a) *Temperature.* Ambient temperature in the range of + 150° to — 65° F. (The 150° F. temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours (see 4.7).

(b) *Relative humidity.* Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours (see 4.8).

3.5 **Bounce and shock.** The equipment shall meet full specification requirements after being subjected to the bounce test (4.6) and the shock test (4.5).

3.6 **Preproduction samples.** The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract (see 6.1 (c) (1)).

3.7 **Lubrication.** All moving parts, except bearings shall be thoroughly lubricated with medium motor oil (S.A.E. No. 30) during assembly.

3.8 **Standard parts.** AN and MIL–Standard parts shall be used whenever they are suitable for the purpose and shall be identified by their part number. Commercial utility parts, such as screws, bolts, nuts, cotter pins, etc., may be used provided they have suitable properties and are replaceable by the AN and MIL part numbers and the part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN and MIL part is in effect on the date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.9 **Identification of product.** Equipment, assemblies and parts shall be identified in accordance with MIL–STD–130.

3.10 **Workmanship.** The item shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.1 Construction.

3.7 Lubrication.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Contractor responsibility.** Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Classification of inspection.** Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (see 4.3).

(b) Procurement inspection (procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.4 and 4.12).

(1) Procurement inspection of items before preparation for delivery (see 4.4).

(2) Procurement inspection of preparation for delivery (see 4.12 and section 5).

4.3 **Preproduction inspection.** This inspection will be performed by the Government unless otherwise specified in the contract (see 6.1). It shall consist of the preproduction inspection specified in table I and the group A and group B inspection specified in tables II and III respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements.

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TABLE I. *Preproduction inspection*

Inspection (For additional preproduction inspection see 4.3)	Requirement paragraph	Inspection paragraph
Shock	3.5	4.5
Bounce	3.5	4.6
Temperature	3.4	4.7
Relative humidity	3.4	4.8

4.4 Procurement inspection of items before preparation for delivery. The contractor shall perform the inspection specified by (a) through (c) below, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government — at its discretion — may perform all or any part of the inspection specified in (a), to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection). Also see 6.3.)

(a) Procurement inspection shall consist of group A and B inspection as specified in 4.4.1 and 4.4.2.

(b) When Standard MIL–STD–105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when it is considered more practical to select separate samples for group B inspection.

4.4.1 *Group A inspection.* This inspection (including sampling) shall conform to table II and Standard MIL–STD–105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (see 4.11) shall be last.

TABLE II. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical	3.10	4.10	1%	4%
Operational	3.2	4.11	1%	

4.4.2 *Group B inspection.* This inspection (including sampling) shall conform to table III and the appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. Disposition of nonconforming product (sample unit and inspection lots) shall be in accordance with 4.4.3 and the requirements of Standard MIL–STD–105 for disposition of rejected product. The AQL shall be 6.5 percent defective and the inspection level shall be L–7 for normal and tightened inspection and L–5 for reduced inspection.

TABLE III. *Group B inspection of equipment*

Inspection	Requirement paragraph	Inspection paragraph	AQL
Interchangeability	3.3	4.9	6.5%

4.4.3 *Disposition of nonconforming product.* When defective sample units or failed inspection lots are submitted for acceptance, such

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product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 **Shock test.** The equipment shall be dropped 3 times with its longest dimension parallel to the floor from a height of 4 feet onto a 2-inch fir floor backed by concrete or a rigid steel frame. Upon completion of this test the equipment shall function properly and meet all the requirements of this specification.

4.6 **Bounce test.** The equipment shall be placed on the table of the Package Tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. During each ½-hour period of the test, the equipment shall rest on a different face.

4.7 **Temperature test for equipment.** The equipment shall be subjected to the temperature cycle shown on Standard MIL–STD–169. Measurements shall be taken as follows:

Paragraph	*Step*	*Measurement*	*Test results*
4.11	3 and 7	Operation	Full specification requirements

4.8 **Moisture resistance test for equipment.** The equipment shall be subjected to two cycles of the moisture resistance test shown on Standard MIL–STD–170 after which the equipment shall meet full specification requirements.

4.9 **Inspection for interchangeability.** The dimensions that affect the mating of parts on the RL–27–( ) shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.3. When a dimension that affects mating or parts is not within design limits, it shall be considered a major defect.

4.10 **Visual and mechanical inspection.** The equipment shall be examined for the defects listed in table IV.

TABLE IV. *Classification of visual and mechanical defects*

Classification	Defect
Major	Rollpin missing or improperly set.
	Crank missing, damaged or broken.
	Axle improperly assembled.
	Washer missing.
	Bushing assembly missing or improper fit.
	Marking omitted or incorrect.
	Axle assembly does not operate smoothly and freely.
Minor	Improper finish.
	Finish damaged or chipped.

4.11 **Operational inspection.** The axle RL–27–( ) completely assembled shall have both Reel DR–4 and Wire Reel RL–159/U mounted on it, one at a time, and shall be used to reel and unreel cable making a minimum of 5 turns of the crank for each test to determine compliance with the requirements of 3.2.

4.12 **Procurement inspection of preparation for delivery.** The contractor shall perform the inspection specified by 4.12.1 and 4.12.2 to demonstrate compliance with specified requirements. However, this does not relieve the contractor of responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government — at its discretion — may perform all or any part of the inspection specified in 4.12.1 and 4.12.2, to verify the contractor's compliance

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with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. Also see 6.2.) When Standard MIL–STD–105 specifies action by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.12.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL–P–116. Classification of defects shall be as shown on table V.

TABLE V. *Preservation and packaging and marking thereof*

Classification	Defect
Major	1. Use of improper or defective material.
	2. Quantity in unit package not as specified.
	3. Incorrect packaging method applied.
	4. Cushioning or padding omitted.
	5. Cushioning or padding inadequate for the protection of the barrier material from projections, sharp edges, or other similar features of the item.
	6. Cushioning inadequate for the physical and mechanical protection of the item.
	7. Stock number omitted, incorrect, or illegible.
	8. Nomenclature omitted, incorrect, or illegible.
	9. Marking of quantity of items in package omitted, incorrect or illegible.
Minor	1. Item not properly blocked or braced within the unit package to prevent movement.
	2. Conforming wraps are not snug-fitting and contain voids.
	3. Any item of marking information other than 7 thru 9 omitted, incorrect, or illegible.

4.12.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to appendix to Standard MIL–STD–105. Inspection level L–8 shall be used for normal inspection and L–6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table VI.

4.12.2.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.12.2.2 *Disposition of nonconforming product.* Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL–STD–105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.13 Rough handling test (preparation for delivery). When rough handling test is required by the contract (see 6.1(c)), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling: 4.11 Operational.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Axle RL–27–( ) shall be preserved packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and as described herein.

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TABLE VI. *Packing, and marking thereof*

Classification	Defect
Major	1. Use of improper or defective material.
	2. Gross weight in excess of specified amount.
	3. Box closure not as specified.
	4. Type, grade, class, and style of the shipping container not as specified.
	5. Strapping omitted.
	6. Strapping inadequate or incorrectly applied.
	7. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
	8. Shipping documents or packing list omitted.
	9. Stock number omitted, incorrect, or illegible.
	10. Nomenclature omitted, incorrect, or illegible.
	11. Marking of quantity of items in pack omitted, incorrect, or illegible.
	12. Destination marking omitted, incorrect, or illegible.
	13. Service designation (color marking) omitted.
	14. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
	15. Overseas code marking omitted, incorrect, or illegible.
Minor	1. Unsealed carton.
	2. Defective taping or sealing of carton.
	3. Any other box defect which may be considered minor by definition of Standard MIL–STD–105.
	4. Any item of required marking information other than 9 thru 15 listed under major detfects omitted, incorrect, or illegible.

5.1.1.1 *Axle RL–27–( )*. Each axle shall be packaged Method IC–5 as follows: Secure the handle (crank) to the axle by tying with type II cotton twine, conforming to Specification T–T–871. Cushion all sharp projections of the bundle with pads fabricated of type II, class I, PPP–P–291. Secure cushioning with class 2 gummed paper tape, Specification UU–T–111. Place the cushioned axle with handle within a close-fitting five panel folder class 2 W6c conforming to Specification PPP–B–636. Seal all seams, corners, and joints with water resistant, pressure-sensitive tape, conforming to Specification PPP–T–76.

5.1.2 *Level C.* Axle RL–27–( ) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* Axle RL–27–( ) shall be packed for shipment as follows: Place a quantity of packaged axles, (see 5.1.1.1) within a close-fitting wood, nailed and lock-corner boxes, wood wirebound boxes, wood-cleated fiberboard boxes, or wood-cleated plywood boxes, conforming to Specification PPP–B–621, PPP–B–585, PPP–B–591, or PPP–B–601 respectively. Boxes shall be overseas class or type. The weight of box with contents shall not exceed 200 pounds. Box closure shall be in conformance with the applicable box specification or appendix thereto.

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.2.2 *Level B.* Axle RL–27–( ) shall be packed for shipment as specified in 5.2.1 except that boxes shall be domestic class or type. No strapping required.

5.2.3 *Level C.* Axle RL–27–( ) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4, and Consolidated Freight Classification Rules 21, for rail shipment, National Motor Freight Classification No. A–3 and No. 14 for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of trans-

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portation employed at lowest transportation rate.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (level A, level B, or level C).

(c) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample (see 3.6).

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon—including rough handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on Drawing SC–C–33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction (pack(s).

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) If preproduction inspection will be performed by a nongovernmental agency (see 4.3).

6.2 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: RL–27–W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.3 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) *Type A.* The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by MIL–STD–105.

(b) *Type B.* That inspection set forth in Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Standard MIL–STD–105.

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(c) *Type C.* A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Signal Corps
Navy—Bureau of Yards and Docks
Air Force—Memphis Air Force Depot

Preparing activity:

Army—Signal Corps

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PROCUREMENT DATA DIVISION
PRODUCTION AND SPECIFICATION ENGINEERING DEPARTMENT
U.S. ARMY ELECTRONICS MATERIEL SUPPORT AGENCY
FORT MONMOUTH, NEW JERSEY

Date: 12 December 1962

ADDENDUM NUMBER 1

To Accompany: Specification MIL-A-11928B

For: AXLE RL-27( ) (FOR LAYING FIELD WIRE)

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1. Paragraph 3.1 Construction.- Delete the last sentence in this paragraph.

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